

Applied Technology School

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Unit One

Exponents and logarithms

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Permutations and combinations and binomial theorem

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Unit One

Exponents and logarithms

Dear student, by the end of studying this unit, you should have the following abilities and knowledge

- ❖ To know the exponential function.
- ❖ To know the laws of the exponents.
- ❖ To solve exponential equations.
- ❖ To know the logarithmic function.
- ❖ To convert the function from exponential form to logarithmic form and vice versa.
- ❖ To know the laws of r of logarithms.
- ❖ To solve the logarithmic equation.
- ❖ To solve problems that include applying the laws of logarithms.
- ❖ To find the value of the logarithm using the calculator.
- ❖ Solve some exponential equations using the logarithm
- ❖ To use a calculator, to solve some exponential equations using the logarithm

Lesson (1)

Non negative integer powers

You have studied and learned that :

$$\diamond X \times X \times X \times X \dots\dots\dots = X^n \quad \text{where } x \in \mathbb{R}, n \in \mathbb{Z}^+$$

$$\diamond X^0 = 1 \quad x \in \mathbb{R} - \{0\}$$

Laws of indices

$$(1) a^m \times a^n = a^{m+n}$$

$$(2) \frac{a^m}{a^n} = a^{m-n}$$

$$(3) (a^n)^m = a^{n \cdot m}$$

$$(4) (ab)^m = a^m b^m$$

$$(5) \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

Example 1

Put in simplest form $3^2 \times 3^5$

Solution

$$3^2 \times 3^5 = 3^7 = 2187$$

Example 2

Put in simplest form $5^2 + 5^2$

Solution

$$5^2 + 5^2 = 25 + 25 = 50$$

Example 3

Put in simplest form $3^4 \times 2^4$

Solution

$$3^4 \times 2^4 = 6^4 = 1296$$

Example 4

Put in simplest form $(2x)^0 \quad x \neq 0$

Solution

$$(2x)^0 = 1$$

Example 5

Put in simplest form $4^x + 4^x + 4^x + 4^x$

Solution

$$\begin{aligned} 4^x + 4^x + 4^x + 4^x &= 4^x (1 + 1 + 1 + 1) \\ &= 4^x (4) = 4^{x+1} \end{aligned}$$

Example 6

Put in simplest form $(3^2)^3$

Solution

$$(3^2)^3 = (3)^6 = 729$$

Exercise (1)

Put in the simplest form using laws of indices

(1) $5^3 \times 5^3$

(2) $4^5 \div 4^4$

(3) $(5y)^0 \quad y \neq 0$

(4) $(x^3)^4 \div x^2$

(5) Fifth of 5^4

Lesson(2)

Negative integer powers indices

$$x^{-n} = \frac{1}{x^n} \quad x \in \mathbb{R} - \{0\}, \quad n \in \mathbb{Z}^+$$

Notice

All of the laws you have studied in non-negative integer powers are true in negative integer powers

Example 1

Put in simplest form $2^4 \times 2^{-2}$

Solution

$$2^4 \times 2^{-2} = 2^{4+(-2)} = 2^2 = 4$$

Example 2

Put in simplest form $(3^2)^{-2}$

Solution

$$(3^2)^{-2} = 3^{-4} = \frac{1}{3^4} = \frac{1}{81}$$

Example 3

Put in the simplest form $\frac{6^{-5} \times 6^2}{6^2}$

Solution

$$\frac{6^{-5} \times 6^2}{6^2} = \frac{6^{-2}}{6^2} = 6^0 = 1$$

Example 4

Put in the simplest form $(4^{-3})^5 (4^6)^3$

Solution

$$\begin{aligned}(4^{-3})^5 (4^6)^3 &= (4)^{-15} \times (4)^{18} \\ &= (4)^3 = 64\end{aligned}$$

Example 5

Put in simplest form $\frac{3^{-5} \times 3^2}{3^{-4}}$

Solution

$$\frac{3^{-5} \times 3^2}{3^{-4}} = (3)^{-5+2+4} = (3)^1 = 3$$

Exercise (2)

Put in simplest form using laws of indices

(1) $3^5 \times 3^{-3}$

(2) $(5^3)^{-1}$

(3) $\frac{2^{-4} \times 2^7}{2^3}$

(4) $(3^{-2})^{-2} \times (3^3)^{-1}$

(5) $\frac{3^{-6} \times 3^3}{3^{-5}}$

Lesson (3) Fractional indices

$$\sqrt[n]{x} = x^{\frac{1}{n}}$$

- ❖ If n is an even number : x non is negative real number
- ❖ If n is an odd number : x is a real number

Example 1

Find the value of $16^{\frac{1}{4}}$

Solution

$$16^{\frac{1}{4}} = \sqrt[4]{16} = 2$$

Example 2

Find the value of $-(27)^{\frac{1}{3}}$

Solution

$$-(27)^{\frac{1}{3}} = -\sqrt[3]{27} = -3$$

Example 3

Find the value of $(-243)^{\frac{1}{5}}$

Solution

$$(-243)^{\frac{1}{5}} = \sqrt[5]{-243} = -3$$

Example 4

Find the value of $(-9)^{\frac{1}{2}}$

Solution

$$(-9)^{\frac{1}{2}} = \sqrt{-9} \notin R \quad \text{remember } \sqrt{-9} = 3i$$

$$\text{definition : } x^{\frac{m}{n}} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$$

Example 5

Find the value of $(16)^{\frac{3}{2}}$

Solution

$$(16)^{\frac{3}{2}} = (\sqrt{16})^3 = (4)^3 = 64$$

Example 6 Find the value of $(27)^{\frac{4}{3}}$

Solution

$$(27)^{\frac{4}{3}} = (\sqrt[3]{27})^4 = (3)^4 = 81$$

Example 7

Simplify then find the value of $\frac{10^{2x+1} \times 2^{2x}}{4^x \times 25^{x+\frac{1}{2}}}$ at $x = \frac{1}{2}$

Solution

$$\frac{10^{2x+1} \times 2^{2x}}{4^x \times 25^{x+\frac{1}{2}}} = \frac{2^{2x+1} \times 5^{2x+1} \times 2^{2x}}{2^{2x} \times 5^{2x+1}} = 2^{2x+1}$$

$$\text{at } x = \frac{1}{2} = 2^{2(\frac{1}{2})+1} = 2^2 = 4$$

Exercise (3)

Put in simplest form using laws of indices

(1) $(125)^{\frac{1}{3}}$

(2) $(81)^{\frac{3}{4}}$

(3) $(128)^{\frac{2}{7}}$

(4) $-(343)^{\frac{2}{3}}$

(5) Prove that $\frac{9^{4x+1} \times 4^{2-2x}}{3^{9x+1} \times 48^{1-x}} = 1$

(6) Prove that $\frac{343^{2x-\frac{1}{3}} \times 4^{3x+1}}{4 \times 196^{3x}} = \frac{1}{7}$

(7) Simplify $\frac{5^2 \times 4^x}{10^{2x-1}}$

Lesson (4)

Solving exponential equations

Rules of solving equation

- ❖ If $x^m = x^n$ then $m = n$
- ❖ If $x^n = y^n$, n odd then $x = y$
- ❖ If $x^n = y^n$, $x \neq y$ then $n = 0$

Example 1

If $3^{x-5} = 9$ find x

Solution

$$3^{x-5} = 3^2$$

$$\therefore x - 5 = 2$$

$$\therefore x = 7$$

Example 2

$3^{x-5} = 1$ find x

Solution

$$3^{x-5} = 1$$

$$\therefore x - 5 = 0$$

$$\therefore x = 5$$

Example 3

$$4^{x-1} = 5^{x-1} \text{ find } x$$

Solution

$$\therefore x - 1 = 0$$

$$\therefore x = 1$$

Example 4

$$5^{2x-1} = \frac{1}{125} \text{ find } x$$

Solution

$$\therefore 5^{2x-1} = \frac{1}{5^3}$$

$$\therefore 5^{2x-1} = 5^{-3}$$

$$\therefore 2x - 1 = -3$$

$$\therefore 2x = -2$$

$$\therefore x = -1$$

Example 5

$$\left(\frac{3}{5}\right)^x = \left(\frac{27}{125}\right)^{-1} \text{ find } x$$

Solution

$$\therefore \left(\frac{3}{5}\right)^x = \left(\frac{3}{5}\right)^{-3}$$

$$\therefore x = -3$$

Example 6

$$(27)^{x+2} = \sqrt[3]{9} \text{ find } x$$

Solution

$$\therefore [(27)^{x+2}]^3 = 9$$

$$\therefore [(3^3)^{x+2}]^3 = 3^2$$

$$\therefore (3)^{9x+18} = 3^2$$

$$\therefore 9x + 18 = 2 \quad 9x = -16 \quad x = \frac{-16}{9}$$

Example 7

Solve the equation $3^{x-1} + 3^{x+1} = 90$

Solution

$$3^{x-1} (1 + 3^2) = 90$$

$$3^x (10) = 90$$

$$3^x = 9$$

$$x = 2$$

$$\therefore \text{S.S.} = \{2\}$$

Exercise (4)

(1) Solve the following equations in R

(i) $(x)^{\frac{7}{2}} = 128$ (ii) $(2x + 3)^{\frac{4}{3}} = 81$ (iii) $(x)^{\frac{3}{2}} = 64$

(2) If $(5)^x = 2$ then $(25)^x = \dots \dots$

(3) Solve the following equations in R

(i) $(x)^{\frac{5}{2}} = \frac{1}{32}$ (ii) $\sqrt[3]{(x-1)^5} = 32$ (iii) $(3)^{x+1} = \frac{1}{27}$

(4) Solve the following equations in R

(i) $(2)^{x-3} = (5)^{x-3}$ (ii) $(7)^{x+1} = (3)^{2x+2}$

(5) Solve the equation $3^{x+1} - 3^{x-1} = 72$

Lesson (5)

Applications on exponential equations

Example1

Solve the equation $49^x - 50 (7)^x + 49 = 0$

Solution

$$7^{2x} - 50 (7)^x + 49 = 0$$

$$(7^x - 49) (7^x - 1) = 0$$

$$7^x = 7^2 \quad \parallel \quad 7^x = 1$$

$$\therefore x = 2 \quad \parallel \quad \therefore x = 0$$

$$S.S = \{0, 2\}$$

Example 2

If $f(x) = 5^x$ solve the equation $f(x) - f(3 - x) = 30$

Solution

$$5^x - 5^{3-x} = 30$$

$$5^x - \frac{125}{5^x} = 30$$

$$5^{2x} - 125 = 30 (5^x)$$

$$5^{2x} - 30 (5)^x - 125 = 0$$

$$(5^x - 25) (5^x - 5) = 0$$

$$\begin{array}{l} 5^x = 5^2 \\ \therefore x = 2 \end{array} \quad \parallel \quad \begin{array}{l} 5^x = 5 \\ \therefore x = 1 \end{array}$$

$$S.S = \{1, 2\}$$

Example 3

If $(x)^{\frac{4}{5}} - 3(x)^{\frac{2}{5}} - 4 = 0$ find x

Solution

$$(x)^{\frac{4}{5}} - 3(x)^{\frac{2}{5}} - 4 = 0$$

$$(x^{\frac{2}{5}} - 4) (x^{\frac{2}{5}} - 1) = 0$$

$$\begin{array}{l} x^{\frac{2}{5}} = 4 \\ \therefore x = 4^{\frac{5}{2}} \end{array} \quad \parallel \quad \begin{array}{l} x^{\frac{2}{5}} = -1 \\ \text{refused} \end{array}$$

$$X = \pm 32$$

Exercise (5)

(1) Find solution set of the equation

$$4^x - 2(2)^x - 8 = 0$$

(2) $f(x) = 7^x$ solve the equation $f(x) - f(2 - x) = 50$

(3) $f(x) = 3^x$ solve the equation $f(x) - f(2 - x) = 6$

(4) Solve the equation $2^{2x} - 6(2)^x + 8 = 0$

(5) $(x)^{\frac{4}{3}} - 3(x)^{\frac{2}{3}} - 4 = 0$ find x

Lesson (6)

The logarithmic function and its relationship to the exponential function

The logarithmic function

If $a \in R^+ - \{1\}$ where $y = \log_a x$ (logarithm form)

We can convert it to exponential form $x = a^y$

Notices that No logarithm to negative number

No logarithm to zero

Example1 Solve the equation $\log_x 81 = 4$

Solution

$$x^4 = 81$$

$$x^4 = 3^4$$

$$x = 3$$

Example 2

Solve the equation $\log_5 125 = x$

Solution

$$5^x = 125$$

$$5^x = 5^3$$

$$x = 3$$

Example 3

Solve the equation $\log_2 (x^2 + 2x) = 3$

Solution

$$2^3 = x^2 + 2x$$

$$x^2 + 2x - 8 = 0$$

$$(x - 2)(x + 4) = 0$$

$$x = 2 \quad \text{or} \quad x = -4$$

$$\therefore \text{s.s} = \{ 2, -4 \}$$

Exercise (6)

(1) Solve the equation $\log_3 (2x - 5) = 1$

(2) Solve the equation $\log_x (x + 2) = 2$

(3) Solve the equation $\log_{81} x = \frac{3}{4}$

(4) Solve the equation $\log_x 5x = 2$

(5) Solve the equation $\log_3 27 = x + 2$

(6) Solve the equation $\log_2 (4^x - 2) = x$

Lesson (7) properties of logarithms

(1) $\log_a x_1 \cdot x_2 = \log_a x_1 + \log_a x_2$ for all $x_1, x_2 \in \mathbb{R}^+$.

(2) $\log_a \frac{x_1}{x_2} = \log_a x_1 - \log_a x_2$ for all $x_1, x_2 \in \mathbb{R}^+$.

(3) $\log_a x^n = n \log_a x$ for all $x \in \mathbb{R}^+$, and for all $n \in \mathbb{R}$.

(4) $\log_a a = 1$ for all $a \in \mathbb{R}^+$.

(5) $\log_a 1 = 0$ for all $a \in \mathbb{R}^+ - \{1\}$.

(6) Property of changing the base $\log_x y = \frac{\log_n y}{\log_n x}$

(7) Property of multiplicative inverse $\log_x y = \frac{1}{\log_y x}$

If the base is not mentioned, then it is 10
which is the natural base of logarithms

$$\log 10 = 1$$

$$\log 100 = 2$$

$$\log 1000 = 3$$

$$\log 0.1 = -1$$

$$\log 0.01 = -2$$

$$\log 0.001 = -3$$

Example 1

Simplify $\log 2 + 2 \log 3 + \log 1 - \log 18$

Solution

$$\begin{aligned} &= \log 2 + \log 9 + \log 1 - \log 18 \\ &= \mathbf{Log} \frac{2 \times 9 \times 1}{18} = \mathbf{Log 1 = 0} \end{aligned}$$

Example 2

Prove that $\log \frac{170}{7} - \log \frac{18}{35} + \log \frac{36}{27} = 2$

Solution

$$\mathbf{Log} \frac{170 \times 35 \times 36}{7 \times 18 \times 27} = \mathbf{Log 100 = 2}$$

Example 3

Prove that $\log 125 + 2 \log 27 - \frac{3}{2} \log 100 = 3 \log 4.5$

Solution

$$\text{L.H.S} = \mathbf{Log} \frac{125 \times (27)^2}{(100)^{\frac{3}{2}}} = \mathbf{Log} \frac{729}{8}$$

$$\text{R.H.S} = \mathbf{Log} (4.5)^3 = \mathbf{Log} \frac{729}{8}$$

Two sides are equal

Exercise (7)

- (1) Simplify to simplest form $\log 2 + \log 5$
- (2) Simplify to simplest form $\log_5 15 - \log_5 3$
- (3) Simplify to simplest form $\log 54 - 3\log 3 - \log 2$
- (4) Simplify to simplest form $\log_{abc} a + \log_{abc} b + \log_{abc} c$
- (5) Prove that $\log_4 38 - \log_4 42 + \log_4 56 - \log_4 19 + \log_4 24 = 3$
- (6) Prove that $\frac{2 \log 9 \times \log 8}{3 \log 2 \times \log 3} = 4$

Lesson (8)

Solving logarithmic equation

Example 1

Solve the equation $\log_2 x = \log_x 2$

Solution

$$\log_2 x = \frac{\log_2 2}{\log_2 x}$$

$$\log_2 x = \frac{1}{\log_2 x}$$

$$(\log_2 x)^2 = 1$$

$$\log_2 x = 1 \quad \log_2 x = -1$$

$$x = 2^1 \quad x = 2^{-1}$$

$$x = 2 \quad x = \frac{1}{2}$$

$$\therefore \text{s.s} = \left\{ 2, \frac{1}{2} \right\}$$

Example 2

Solve the equation $\log (x^2 + 9x) = 1$

Solution

$$10 = x^2 + 9x$$

$$x^2 + 9x - 10 = 0$$

$$(x - 1)(x + 10) = 0$$

$$x = 1 \quad \text{or} \quad x = -10$$

$$\therefore \text{s.s} = \{ 1, -10 \}$$

Example 3

Solve the equation

$$\log_2 (x^2 + 6x + 9) - \log_2 (x - 1) = \log_5 625$$

Solution

$$\log_2 \frac{x^2 + 6x + 9}{x - 1} = \log_5 5^4 = 4 \log_5 5$$

$$\log_2 \frac{x^2 + 6x + 9}{x - 1} = 4$$

$$\frac{x^2 + 6x + 9}{x - 1} = 16$$

$$x^2 + 6x + 9 = 16x - 16$$

$$x^2 - 10x + 25 = 0$$

$$(x - 5)(x - 5) = 0$$

$$x = 5$$

$$\therefore \text{s.s} = \{ 5 \}$$

Exercise (8)

- (1) Solve the equation $\log_2 x + \log_x 2 = 2$
- (2) Solve the equation $\log_2 x + \log_2 (x + 12) = 3$
- (3) Solve the equation $\log (x - 1)^3 + 3 \log (x - 3) = 8$
- (4) Solve the equation $\log_2 2x + \log_x 2 = 3$
- (5) Solve the equation $\log (x + 8) - \log (x - 1) = 1$
- (6) Solve the equation $\log_3 (x + 6) = 2 \log_3 x$

Lesson (9)

Using a calculator to find logarithms

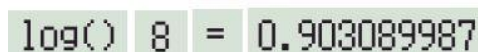
The calculator can be used to find the logarithms value

Example 1 Find the value of $\log_2 4$



$\log_2(4) = 2$

Example 1 Find the value of $\log 8$



$\log(8) = 0.903089987$

Use a calculator to find :

$$\log_3 12$$

$$\text{Log}_3 24$$

$$\text{Log } 128$$

$$\log_5 125$$

$$\text{Log } 100$$

$$\text{Log } 1000$$

$$\log_5 500$$

Exercise (9)

Use a calculator to find :

$$\log_3 81$$

$$\text{Log}_2 \frac{1}{8}$$

$$\text{Log}_{\sqrt{7}} 343$$

$$\text{Log}_2 4\sqrt{2}$$

$$\text{Log}_7 1$$

$$\text{Log}_2 16$$

Lesson (10)

Solving Exponential Equations Using Logarithms

Example 1

Solve the equation $2^{x-1} = 3^{x+2}$

Solution

$$\log 2^{x-1} = \log 3^{x+2}$$

$$(x-1)\log 2 = (x+2)\log 3$$

$$x\log 2 - \log 2 = x\log 3 + 2\log 3$$

$$x\log 2 - x\log 3 = 2\log 3 - \log 2$$

$$x\log 2 - x\log 3 = 2\log 3 - \log 2$$

$$x(\log 2 - \log 3) = 2\log 3 - \log 2$$

$$x = \frac{2\log 3 - \log 2}{\log 2 - \log 3} = 2.66$$

Example 1

If $3^{2x-5} = 5^{x+2}$ Find x

Solution

$$(2x - 5) \log 3 = (x + 2) \log 5$$

$$2x \log 3 - 5 \log 3 = x \log 5 + 2 \log 5$$

$$2x \log 3 - x \log 5 = 5 \log 3 + 2 \log 5$$

$$x(2 \log 3 - \log 5) = 5 \log 3 + 2 \log 5$$

$$x = \frac{5 \log 3 + 2 \log 5}{2 \log 3 - \log 5} = 14.82$$

Exercise (10)

(1) If $(x + 4)^{\frac{2}{3}} = 6.123$ Find x

(2) If $7^{3x-2} = 5$ Find x

(3) If $7^{x+1} = 5^{x-2}$ Find x

(4) If $7^{x-2} = 4^{x+3}$ Find x

(5) If $7^{7-2x} = 13.4$ Find x

Unit test

The first question :

- (a) Solve the equation $3^{x+1} + 3^{x-1} = 90$
(b) Solve the equation $\log_2 (x^2 + 2x) = 3$

The second question :

- (a) Put in simplest form $\frac{3^{-5} \times 3^2}{3^{-4}}$
(b) Solve the equation $\log 2 + 2 \log 3 + \log 1 - \log 18$

The third question :

- (a) Put in simplest form $\frac{5^2 \times 4^x}{10^{2x-1}}$
(b) Solve the equation $\log (x-1)^3 + 3 \log (x-3) = 8$

The fourth question :

(a) Solve the equation $\log_2 x + \log_x 2 = 2$

(b) Solve the equation $4^x + 2(2)^x - 8 = 0$

The fifth question :

(a) $(x)^{\frac{4}{3}} - 3(x)^{\frac{2}{3}} - 4 = 0$ find x

(b) Solve the equation If $7^{3x-2} = 5$ Find x

Evaluation

Unit Two

Permutations and Combinations and Binomial Theorem

Dear student, by the end of studying this unit, you should have the following abilities and knowledge

- ❖ To know the principle of counting and its simple applications.
- ❖ To know an introduction to permutations and combinations.
- ❖ To use a calculator to calculate permutations and combinations.
- ❖ To know the binomial theorem with a positive true power.
- ❖ To derive the general term in Binomial Expansion.
- ❖ Binomial Expansion.
- ❖ That there is a coefficient of any term in
- ❖ To find the a middle term free of x in Binomial Expansion
- ❖ To find the a middle term in Binomial Expansion.
- ❖ To deduce the ratio between any term and its previous term in Binomial expansion

Lesson (1) Counting principle

Basic Counting Principle

If: Number of ways to do an action = m

The number of ways to perform a second work = k

The number of ways to perform a third work = n and so on

Then the number of ways to do these actions together = $m \times k \times n \dots$

Example 1:

By how many ways it is possible to choose a boy from a group of three boys and a girl from a group of two girls?

Solution:

Number of ways = $3 \times 2 = 6$ ways.

Example 2:

By how many ways it is possible to choose a uniform which consists of shirt and trousers from 5 shirts and 3 trousers?

Solution:

Number of ways = $5 \times 3 = 15$ ways.

Example 3

How many 3 digit numbers can be formed from the digits {1, 2, 3, 4 }?

Solution

Number of ways = $4 \times 4 \times 4 = 64$ ways

units	tens	H.
4	4	4

Example 4

How many 3 digits can be formed from the numbers {0, 1, 2, 3, 4 }?

Solution

Number of ways = $5 \times 5 \times 4 = 100$ ways

Example 5: (additional condition)

By how many ways it is possible to form different 3-digit number from the set {1, 2, 3, 4}?

Solution:

units	tens	H.
4	3	2

Number of ways = $4 \times 3 \times 2 = 24$ ways.

One of the digits takes units place and not repeated in the other places

One of the 3 remainder digits takes tens place and not use again.

The last 2 digits takes hundreds place.

Example 6: (additional condition)

By how many ways it is possible to form 3-digit different number from the set {0, 1, 2, 3, 4}?

Solution:

units	tens	H.
3	4	4

Number of ways = $3 \times 4 \times 4 = 48$ ways.

We cannot put the zero in the hundreds place.

Number of ways to choose a digit in the hundreds place = 4

Number of ways to choose a digit in the tens place = 4

Number of ways to choose a digit in the units place = 3

Exercise (1)

- (1) How many 2 digits number can be formed from the numbers {5,3, 0, 2 }?
- (2) How many ways can four students sit on four chairs in row?
- (3) How many odd numbers of three different digits are taken from the numbers {2,3, 6, 8 }?
- (4) How many odd numbers of four different digits are taken from the numbers {2,3, 4, 5 } so that the number in tens is an even number
- (5) How many 3-digit numbers are taken from the elements {2,3, 6, 8 }?
- (6) How many 3-digit numbers are taken from the elements {2,3, 6, 8 } so that the number in units is 6 ?

Lesson (2)

Factorial of the number & permutation

Factorial of a positive integer number (n) is write as $n!$

$$n! = n(n-1)(n-2)\dots\dots\dots\times 3\times 2\times 1$$

Notes:

$$0! = 1 \quad 1! = 1 \quad n! = n(n-1)!$$

Number of sitting down of (n) people in a one row = $n!$

Number of sitting down of (n) people in a circle = $(n-1)!$

Example 1:

Find

(a) $\frac{10!}{8!}$

(b) if $n! = 120$ then find the value of n

Solution:

(a) $\frac{10!}{8!} = \frac{10 \times 9 \times 8!}{8!} = 90$

$$(b) \quad \underline{n} = 120 \quad \longrightarrow \quad \underline{n} = \underline{5} \quad \longrightarrow \quad n = 5$$

Example 2:

Find the solution set of: $\frac{\underline{n}}{\underline{n-2}} = 30$

Solution:

$$\frac{n(n-1) \underline{n-2}}{\underline{n-2}} = 30 \quad \longrightarrow \quad n(n-1) = 6 \times 5 \quad n = 6$$

Permutations

How many 3-digit numbers can be formed from the set { 2, 3, 5 }

Number of numbers = $3 \times 2 \times 1 = 6$ numbers.

Every number of those numbers is called permutation and is written as nPr

where ${}^n P_r = n(n-1)(n-2) \dots (n-r+1) \quad r \leq n, r \in \mathbb{N} \text{ and } n \in \mathbb{Z}^+$

Notes:

(a) ${}^n P_{\text{zero}} = 1$ (b) ${}^n P_r = \frac{\underline{n}}{\underline{n-r}}$

Example 1:

- ${}^6 P_3 = 6 \times 5 \times 4 = 120$
- ${}^7 P_5 = 7 \times 6 \times 5 \times 4 \times 3 = 2520$
- ${}^4 P_4 = 4 \times 3 \times 2 \times 1 = 24$

Example 2:

Find the number of ways to 5 students to sit 7 seats in a row.

Solution:

$$\text{Number of ways} = {}^7 P_5 = 7 \times 6 \times 5 \times 4 \times 3 = 2520$$

We can use the calculator:



Example (3)

Find the number of different ways 7 students sit in 7 seats in a circle

Solution
$$= 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

Example (4) ${}^7P_r = 840$, find $\underline{r-4}$

Solution

$${}^7P_r = 840 = {}^7P_4$$

$$r = 4$$

$$\underline{r-4} = \underline{4-4} = \underline{0} = 1$$

Exercise (2)

- (1) By how many ways it is possible to form 2-digit different number from the set $\{3, 4, 5, 6\}$?
- (2) How many ways can 7 children arrange themselves such that they sit around a circular seats?
- (3) How many ways can a president and vice president be selected from a 12-member committee?
- (4) Find the value of n which satisfies:
- (a) $\underline{\hspace{1cm}} n = 24$
- (b) $\frac{\underline{\hspace{1cm}} n + 1}{\underline{\hspace{1cm}} n - 1} = 42$
- (c) ${}^{15}P_n = 2730$
- (5) If: ${}^9P_{r-1} = 504$ then find the value of: $\underline{\hspace{1cm}} r + 1$

Lesson (3) Combinations

Definition:

We denote to number of combinations each of them (r) objects and chosen from (n) elements by nC_r where $r \leq n$, $r \in \mathbb{N}$, $n \in \mathbb{Z}^+$

The rules of combinations:

For all $r \leq n$, $r \in \mathbb{Z}^+$, $n \in \mathbb{Z}^+$

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

- If ${}^nC_x = {}^nC_y$, then: $x = y$ or $x + y = n$
- ${}^nC_r = {}^nC_{n-r}$
- ${}^nC_n = {}^nC_{\text{zero}} = 1$

$$\bullet \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{n}$$

Example (1)

$${}^nC_{n-2} = 36 \text{ find } n$$

Solution

$${}^nC_{n-2} = {}^nC_2 = 36 = {}^9C_2$$

$$n = 9$$

Example (2)

$${}^{21}C_{4n-7} = {}^{21}C_{3n} \text{ find } n$$

Solution

$$4n - 7 = 3n$$

$$n = 7$$

$$4n - 7 + 3n = 21$$

$$7n = 28 \quad n = 4$$

Example (3)

$${}^7C_r = {}^7C_{3r-5}, \quad {}^nC_r : {}^nC_{r-1} = 8:3 \text{ find } n, r$$

Solution

$${}^7C_r = {}^7C_{3r-5}$$

$$4r - 5 = 7$$

$$4r = 12$$

$$r = 3$$

$${}^nC_r : {}^nC_{r-1} = 8:3$$

$${}^nC_3 : {}^nC_2 = 8:3$$

$$\frac{n-3+1}{3} = \frac{8}{3}$$

$$n - 3 + 1 = 8 \quad n = 10$$

Example (4)

Using a calculator find the value of ${}^5C_4 + {}^7C_2$

0	shift	÷	ξ	+	√	shift	÷	√	=
---	-------	---	---	---	---	-------	---	---	---

$${}^5C_4 + {}^7C_2 = 26$$

Exercise (3)

- (1) ${}^{13}C_r : {}^{13}C_{r+1} = 9 : 5$ find r
- (2) ${}^nC_{n-3} = 120$ find n
- (3) ${}^{25}C_{3n-5} = {}^{25}C_{2n}$ find n
- (4) Using a calculator find the value of ${}^{17}C_9 - {}^{17}C_4$
- (5) In how many ways can a team of 4 people be chosen from a group of 9 people .
- (6) A classroom has 10 boys, 8 girls. In how many ways can a five-activity committee be formed, consisting of 3 boys , 2 girls ?

Lesson (4)

Applications on permutations and

Example (1)

$${}^8P_{r+1} = 336 \quad \text{find } {}^{2r}C_4$$

Solution

$${}^8P_{r+1} = 336 = {}^8P_3$$

$$r + 1 = 3 \quad r = 2$$

$${}^{2r}C_4 = {}^4C_4 = 1$$

Example (2)

$$\underline{3n - 7} = 120 \quad \text{find } {}^nC_{n-1}$$

Solution

$$\underline{3n - 7} = 120 \quad 3n - 7 = 5 \quad 3n = 12 \quad n = 4$$

$${}^nC_{n-1} = {}^4C_3 = 4$$

Example (3)

$${}^{2n-1}C_3 = 84 \text{ find } \underline{n - 5}$$

Solution

$${}^{2n-1}C_3 = 84 = {}^9C_3 \quad 2n - 1 = 9 \quad 2n = 10 \quad n = 5$$

$$\underline{n - 5} = \underline{0} = 1$$

Example (4)

$${}^np_2 = 42, \underline{n - r} = 120 \text{ find } {}^nC_{r+3}$$

Solution

$${}^np_2 = 42 = {}^7p_2 \quad \underline{n - r} = 120$$

$$n = 7$$

$$7 - r = 5$$

$$r = 2$$

$${}^nC_5 = {}^7C_5 = 21$$

Example (5)

$${}^{n-m}P_3 = 210, {}^{n+m}C_4 = 715 \text{ find } m \times n$$

Solution

$${}^{n-m}P_3 = 210 = {}^7P_3$$

$${}^{n+m}C_4 = 715 = {}^{13}C_4$$

$$n - m = 7 \longrightarrow \textcircled{1}$$

$$n + m = 13 \longrightarrow \textcircled{2}$$

From 1, 2 we get $n = 10, m = 3$

$$\therefore m \times n = 30$$

Exercise (4)

(1) ${}^n P_r = 120$, ${}^n C_r = 120$ find ${}^{n-2} C_{2r}$

(2) ${}^m C_n = {}^m C_{2n+1}$, $\underline{\quad 2n \quad} = 24$ find m

(3) ${}^n P_5 = 6720$, ${}^n C_{r-1} = 56$ find n, r

(4) $\underline{\quad n-2 \quad} = 24$ find ${}^8 C_n$

(5) ${}^x P_2 = 120$, ${}^{n+1} C_{n-1} = 36$ find $\underline{\quad 3n-4x \quad}$

Lesson (5)

Binomial theorem

The binomial theorem:

$$(x + b)^n = x^n + {}^nC_1 b^1 x^{n-1} + {}^nC_2 b^2 x^{n-2} + {}^nC_3 b^3 x^{n-3} \dots + b^n$$

$$(x + b)^n = x^n - {}^nC_1 b^1 x^{n-1} + {}^nC_2 b^2 x^{n-2} - {}^nC_3 b^3 x^{n-3} \dots + (-b)^n$$

Notice on expansion

1- number of terms = $n+1$

$$2- {}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5 = 2^5$$

$$3- {}^5C_0 - {}^5C_1 + {}^5C_2 - {}^5C_3 + {}^5C_4 - {}^5C_5 = \text{zero}$$

Example 1

Find the expansion of $(1 + 2x)^4$

Solution

$$(1 + 2x)^4 = 1 + {}^4C_1 2x + {}^4C_2 (2x)^2 + {}^4C_3 (2x)^3 + {}^4C_4 (2x)^4$$

$$= 1 + 8x + 24x^2 + 32x^3 + 16x^4$$

Example 2

Find the expansion of $(x - 1)^5$

Solution

$$\begin{aligned}(x - 1)^5 &= x^5 - {}^5C_1 x^4 + {}^5C_2 x^3 - {}^5C_3 x^2 + {}^5C_4 x - 1 \\ &= x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 1\end{aligned}$$

Example 3

Find the expansion of $(2x + 3y)^4$

Solution

$$\begin{aligned}&= (2x)^4 + {}^4C_1 (3y)^1 (2x)^3 + {}^4C_2 (3y)^2 (2x)^2 + {}^4C_3 (3y)^3 (2x) + (3y)^4 \\ &= 16x^4 + 96x^3y + 216x^2y^2 + 216xy^3 + 81y^4\end{aligned}$$

Example 4

Find the sum of coefficient of all terms in the exp. $(1 + 2x)^6$

Solution

To find the sum of coefficient of all terms in the exp. $(1 + 2x)^6$ Put $x = 1$

$$\text{the sum of coefficient of all terms} = (1 + 2)^6 = 729$$

Exercise (5)

- (1) Find the expansion of $(1 + 3x)^6$ Then, deduce the value of the sum of the coefficients of the terms the of expansion
- (2) Find the expansion of $(2 + x)^5$ Then, deduce the value of the sum of the coefficients of the terms the of expansion
- (3) Find the expansion of $(1 - 2x)^7$ Then, deduce the value of the sum of the coefficients of the terms the of expansion
- (4) Find the expansion of $(3 - 2x)^5$ Then, deduce the value of the sum of the coefficients of the terms the of expansion

Lesson (6)

The general term in the Expansion

The general term in the expansion: $(x + a)^n$

$$T_{r+1} = {}^nC_r (a)^r (x)^{n-r}$$

Example 1

In $(x + \frac{2}{x})^8$ find the Coefficient of T_6

Solution

$$T_6 = {}^8C_5 (x)^3 \left(\frac{2}{x}\right)^5$$

Put $x=1$

$$\text{Coefficient of } T_6 = {}^8C_5 (1)^3 (2)^5 = 1792$$

Example 2

In $(2x + 3)^7$ if the third term equals the seventh term find value of x

Solution

$$T_3 = {}^7C_2 (3)^2 (2x)^5 = 6048 x^5$$

$$T_7 = {}^7C_6 (3)^6 (2x)^1 = 10206 x$$

$$6048 x^5 = 10206 x$$

$$x^4 = \frac{27}{16}$$

$$x = \pm 1.14$$

Example 3

In the expansion $(1 + x)^n$ according to the ascending power of x if the third term equals $28 x^2$, $T_5 = 1120$

find x , n

Solution

$${}^nC_2 (x)^2 = 28 x^2$$

$${}^nC_2 = 28$$

$$n = 8$$

$${}^nC_4 (x)^4 = 1120$$

$$70 (x)^4 = 1120$$

$$x = \pm 2$$

Exercise (6)

- (1) In the expansion of $(ax^2 + \frac{1}{ax})^{11}$
if coefficient of T_6 = coefficient of T_7 find value of a
- (2) In the expansion of $(x^2 + \frac{m}{x^3})^{15}$
if coefficient of T_5 = 2 coefficient of T_4 find value of x
- (3) In the expansion of $(3 + bx)^9$
if coefficient of T_3 = coefficient of T_4 find value of b
- (4) In the expansion of $(\frac{1}{x} - x^2)^9$
if coefficient of T_5 = 12 coefficient of T_4 find value of x

- (5) In the expansion of $(x + \frac{1}{x})^4$ according to the descending power of x Find T_4
- (6) In the expansion of $(1 + 2x)^{10}$ according to the ascending power of x Find T_5

Lesson (7)

The middle term in the expansion

Notice on The middle term in the expansion $(x + a)^n$

- ❖ If n is even the expansion contains one middle of order $\frac{n+2}{2}$
- ❖ If n is odd the expansion contains two middle of orders $\frac{n+1}{2}$, $\frac{n+3}{2}$

Example 1

Find the middle term in the expansion $(3x - 1)^{10}$

Solution

The order of the two middle terms $\frac{10+2}{2} = 6$

$$T_6 = {}^{10}C_5 (-1)^5 (3x)^5 = -61236 x^5$$

Example 2

If the middle term in the expansion of $(x^2 - \frac{1}{x})^{n+5}$ is T_{12} find n

Solution

The expansion contains one middle is $T_{12} =$

$$\frac{n + 5 + 2}{2} = 12$$

$$n + 7 = 24$$

$$n = 17$$

Example 3

If the number of terms in the expansion $(x + y)^{2n-1}$ is 12 terms find value of n

Solution

$$2n - 1 = 11$$

$$2n = 12$$

$$n = 6$$

Example 4

If the middle term in the expansion of $(3x^2 + \frac{2}{3x})^8 = 17920$ find x

Solution

$$\text{The order of the two middle terms} = \frac{8 + 2}{2} = 5$$

$$T_5 = 17920$$

$$T_5 = {}^8C_4 \left(\frac{2}{3x}\right)^4 (3x^2)^4$$

$$1120 x^4 = 17920$$

$$x^4 = 16$$

$$x = \pm 2$$

Example 5

In the exp. $(\frac{x^3}{2} - \frac{4}{x})^{11}$ according to the descending power of x if the sum of the two middle terms are equal zero find x ?

Solution

$$T_6 + T_7 = \text{zero}$$

$${}^{11}C_5 \left(\frac{x^3}{2}\right)^6 \left(-\frac{4}{x}\right)^5 + {}^{11}C_6 \left(\frac{x^3}{2}\right)^5 \left(-\frac{4}{x}\right)^6 = 0$$

$$-16x^{13} + 128x^9 = 0$$

$$-16x^4 + 128 = 0 \quad x^4 = 8 \quad x = \pm \sqrt[4]{8}$$

Exercise (7)

(1) If the middle term in the expansion of

$$\left(x^2 + \frac{1}{2x}\right)^{10} = \frac{28}{27} \quad \text{find value of } x$$

(2) In the exp. $\left(\frac{x^2}{3} + \frac{3}{x}\right)^{15}$ find the two middle terms

(3) In the exp. $(3x + 2y)^{13}$ if the two middle terms are equal prove that $3x = 2y$?

(4) If the middle term in the expansion of

$$\left(\frac{2a}{3} + \frac{b}{a^2}\right)^{8n} \text{ is } T_9 \text{ find value of } n$$

(5) In the exp. $\left(2x^2 + \frac{1}{2x}\right)^{12}$ find the middle terms

Lesson (8)

The term which contains x^k in the expansion

Example 1

In the expansion $(1 + x)^{10}$ find the coefficient of contain x^3

Solution

$$T_{r+1} = {}^{10}C_r (x)^r$$
$$r = 3$$

The coefficient of contain $x^3 = {}^{10}C_3 = 120$

Example 2

In the expansion $(3 - 2x)^7$ find the coefficient of x^5

Solution

$$T_{r+1} = {}^7C_r (-2x)^r (3)^{7-r}$$

$$= {}^7C_r (-2x)^r (3)^{7-r}$$

$$r = 5$$

$$T_6 = {}^7C_5 (-2x)^5 (3)^2$$

$$\text{The coefficient of } x^5 = {}^7C_5 (-2)^5 (3)^2 = -6048$$

Example 3

In the exp. $(\frac{a}{x^2} + x)^9$ if the free term of x is 672 find a

Solution

$$\begin{aligned} T_{r+1} &= {}^9C_r (x)^r (\frac{a}{x^2})^{9-r} \\ &= {}^9C_r (x)^r (a)^{9-r} (x)^{-18+2r} \end{aligned}$$

$$= {}^9C_r (a)^{9-r} (x)^{-18+3r}$$

$$-18 + 3r = 0$$

$$r = 6$$

$$T_7 = {}^9C_6 (a)^3$$

$${}^9C_6 (a)^3 = 672$$

$$a^3 = 8 \quad a = 2$$

Example 4

In the expansion $(ax + \frac{1}{x})^{10}$ if the free term of x equals the coefficient of T_7 find a

Solution

$$T_{r+1} = {}^{10}C_r (ax)^{10-r} (\frac{1}{x})^r$$

$$= {}^{10}C_r (a)^{10-r} (x)^{10-2r}$$

$$10 - 2r = 0$$

$$r = 5$$

the free term of x is $T_6 =$ coefficient of $T_7 = {}^9C_6 (a)$

$${}^{10}C_5 (a)^5 = {}^{10}C_6 (a)^4$$

$$a = \frac{5}{6}$$

Exercise (8)

(1) In the expansion $(1 + 2x)^{10}$ Find the term include x^4

(2) In the expansion $(x + \frac{1}{x})^{10}$ Find the term free of x

(3) In the expansion $(4x^2 + \frac{1}{2x})^{12}$ Find the term free of x

(4) In the expansion $(2x - \frac{1}{2x^2})^9$ find

(i) The term free of x

(ii) The coefficient of x^3

(iii) Prove that the expansion has no term include x^2

Lesson (9)

The ratio between two consecutive terms
in the expansion of $(ax + by)^n$

$$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \times \left(\frac{by}{ax} \right)$$

Example 1 In the exp. $(3x - 4y)^{12}$ find $\frac{T_5}{T_4}$

Solution

$$\frac{T_5}{T_4} = \frac{12-4+1}{4} \times \left(\frac{-4y}{3x} \right) = \frac{-3y}{x}$$

Example 2

In the exp. $(x^2 + \frac{1}{x})^{12}$ find the ratio between the term free of x and the middle term at $x=1$

Solution

$$\begin{aligned}
 T_{r+1} &= {}^{12}C_r \left(\frac{1}{x}\right)^r (x^2)^{12-r} \\
 &= {}^{12}C_r (x)^{-r} (x)^{24-2r} \\
 &= {}^{12}C_r (x)^{24-3r} \\
 -24 + 3r &= 0 \\
 r &= 8
 \end{aligned}$$

$$T_9 = {}^9C_8$$

$$T_7 = {}^{12}C_6 (x)^6$$

$$\text{At } x=1 \quad \frac{T_9}{T_7} = \frac{15}{112}$$

Example 3

In the exp. $(x^5 + \frac{1}{x})^6$ find the ratio between the term free of x and the coefficient of the middle term

Solution

$$\begin{aligned}
 T_{r+1} &= {}^6C_r \left(\frac{1}{x}\right)^r (x^5)^{6-r} \\
 &= {}^6C_r (x)^{30-6r} \\
 30 - 6r &= 0 \quad r = 5
 \end{aligned}$$

$$T_6 = {}^6C_5 = 6$$

$$\begin{aligned}
 T_4 &= {}^6C_3 (x)^{18} \\
 &= \frac{15}{112}
 \end{aligned}$$

the ratio between the term free of x and the middle term $= \frac{6}{20} = \frac{3}{10}$

Example 4

In the exp. $(\frac{2x}{3} + \frac{3}{2x^2})^{12}$ if the ratio between the middle term and the term which contains x^{-3} is $\frac{7}{9}$ find x

Solution

$$T_{r+1} = {}^{12}C_r \left(\frac{3}{2x^2}\right)^r \left(\frac{2x}{3}\right)^{12-r}$$

$$= {}^{12}C_r (x)^{12-3r} \left(\frac{2}{3}\right)^{12-2r}$$

$$12 - 3r = -3 \quad r = 5$$

T_6 contains x^{-3}

The middle T_7

$$\frac{T_7}{T_6} = \frac{7}{9} \quad \frac{12-6+1}{6} \times \frac{9}{4x^3} = \frac{7}{9}$$

$$x^3 = \frac{27}{8} \quad x = \frac{3}{2}$$

Exercise (9)

- (1) In the expansion $(1 + x)^{27}$ if the ratio between the two middle terms is $3 : 1$ find x
- (2) In the exp. $(3a - 2b)^{11}$ if the ratio between the two middle terms is $-3 : 2$ find $a : b$
- (3) In the exp. $(x + y)^{10}$ find $\frac{T_9}{T_8}$
- (4) In the exp. $(x^2 + \frac{1}{x})^{18}$ find the ratio between the term free of x and the coefficient of the middle term
- (5) In the exp. $(\frac{1}{x} - x^2)^9$ if $T_5 = 12 T_4$ find x

Lesson (10)

Applications on binomial theorem

Example 1

In the expansion $(x^2 + \frac{1}{8x})^{13}$ if $T_4 = T_{11}$ find x

Solution

$${}^{13}C_3 \left(\frac{1}{8x}\right)^3 (x^2)^{10} = {}^{13}C_{10} \left(\frac{1}{8x}\right)^{10} (x^2)^3$$

$${}^{13}C_3 \left(\frac{1}{8}\right)^3 (x)^{17} = {}^{13}C_{10} \left(\frac{1}{8}\right)^{10} (x)^{-4}$$

$$(x)^{21} = \left(\frac{1}{8}\right)^7$$

$$(x)^{21} = \left(\frac{1}{2}\right)^{21} \quad x = \frac{1}{2}$$

Example 2

In the expansion $(1 + 3x + 3x^2 + 3x^3)^{10}$ Find the coefficient of x^4

Solution

$$(1 + 3x + 3x^2 + 3x^3)^{10} \\ = [(1 + x)^3]^{10} = (1 + x)^{30}$$

$$T_{r+1} = {}^{30}C_r (x)^r$$

$$\text{the coefficient of } x^4 = {}^{30}C_4 = 27405$$

Example 3

In the expansion $x^3 (1 + x)^7$ Find the coefficient of x^4

Solution

$$T_{r+1} = x^3 {}^7C_r (x)^r = {}^7C_r (x)^{r+3}$$

$$\text{the coefficient of } x^4 = {}^7C_1 = 7$$

Example 4

In the exp. $(\frac{x^3}{2} - \frac{2}{x^2})^{12}$ Find the fifth term from the end

Solution

$$(-\frac{2}{x^2} + \frac{x^3}{2})^{12} \text{ then find the fifth term}$$

$$T_5 = {}^{12}C_4 (\frac{x^3}{2})^4 (-\frac{2}{x^2})^8$$

$$= {}^{12}C_4 (\frac{1}{2})^4 (x^3)^4 (-2)^8 (x)^{-16}$$

$$= {}^{12}C_4 (\frac{1}{2})^4 (-2)^8 (x)^{-4}$$

$$= {}^{12}C_4 (2)^4 (x)^{-4}$$

$$= 7920 (x)^{-4}$$

Exercise (10)

- (1) In the expansion $(3 - 2x)^7$ Find the coefficient of x^5
- (2) In the expansion $(2x + \frac{1}{x^2})^{15}$ find the term free of x and prove that the expansion has no term include x^5
- (3) In the expansion $(2x - 3)^{15}$ if $13T_3 + 10T_4 + T_5 = 0$ find x
- (4) In the exp. $(x^2 + \frac{1}{x})^{15}$ find the ratio between the term free of x and the sum of coefficient of the two middle terms

(5) In the exp. $(3 + ax)^9$ if coefficient of $T_3 =$ coefficient of T_4 find x

(6) In the exp. $(1 + x)^{27}$ if the ratio between the two middle terms is $3 : 1$ find x

Unit test

The first question :

(a) ${}^9P_r = 504$, find $|r - 1|$

(b) In the expansion $(3 - 2x)^9$ Find the coefficient of x^4

The second question :

(a) In the exp. $(x + \frac{1}{x})^{18}$ find the ratio between the term free of x and the coefficient of the middle term

(b) In the expansion $(2x + \frac{1}{2x^2})^9$

Prove that the expansion has no term include x^2

The third question :

(a) In the exp. $(4x + 5b)^{8n}$ if the middle term is T_9 find n

(b) ${}^n p_r = 720$, ${}^n C_r = 120$ find ${}^{n-2} C_{2r}$

The fourth question :

(a) ${}_n P_{2n} = 24$, ${}^m C_n = {}^m C_{2n+1}$ find m

(b) Find the expansion of $(3 - 2x)^5$ Then, deduce the value of the sum of the coefficients of the terms the of expansion

Evaluation

Applied Technology School